Coupling the Point Model to the Growth Field

Progress since the meeting of 20 October 2026

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Where the coupling stands

- Implemented as agreed in your element: the point model (Klempt et al. 2026) runs at every Gauss point as published, consumption by the living biomass $\sum_i \bar{\phi}_i$, antibiotic in the second nutrient field. A Python version solves the same equations.
- Composition is robust: $\sum_i \bar{\phi}_i$, the shares and c agree between the two codes and change by less than 1 % from 16^3 to 128^3.
- The front term of Eq. 34 needs care: at the paper's r the cell Péclet number is about **31** on 16^3, so a central gradient oscillates.
- With a second-order scheme the seed von Mises stress converges to 6.2×10^{-4} **Pa** (separate example, from 32^3 on).

The coupling agreed on 20 October

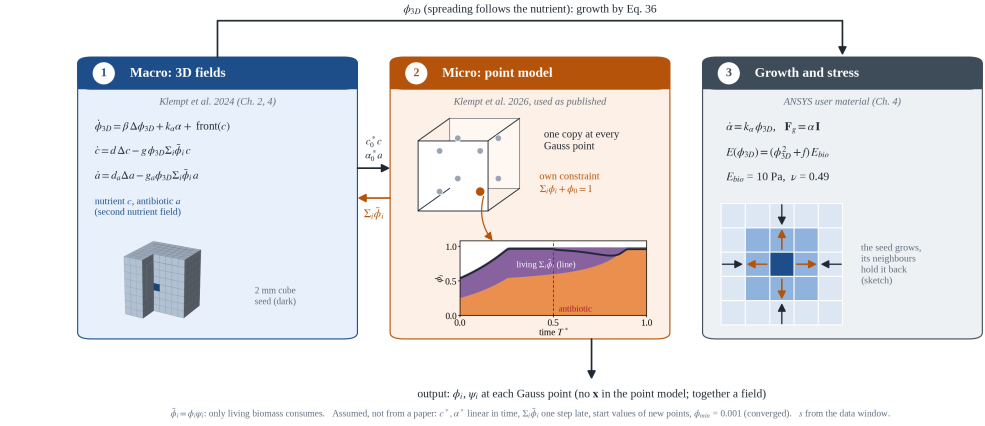
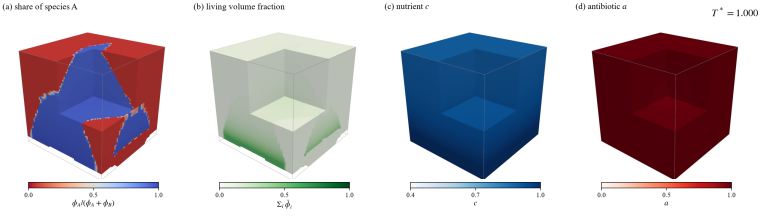


Figure 1: Agreed coupling: the 3D fields of Klempt et al. (2024) pass c^* and α^* to the point model of Klempt et al. (2026) at each Gauss point, which returns $\sum_i \bar{\phi}_i$ for the consumption in Eq. 35.

Separate example with antibiotic

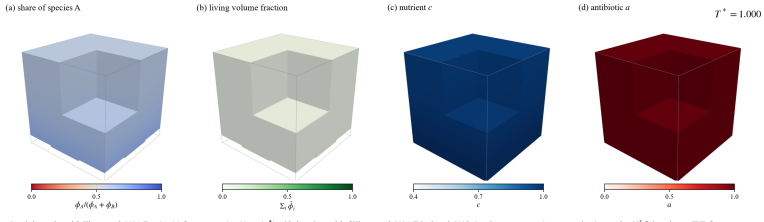


Coupled growth model: Klempt et al. 2024, Eqs. 34–36, front term on ($r = 10$ mm/T^*), with the point model of Klempt et al. 2026 (Fritsch et al. 2025, Case I, true parameters) at every point; 2 mm cube, 64^3 finite volumes, TVD front term.
Separate start: species A in the centre seed, species B in four small blocks at the edges along z (lol, hl > 0.75 mm). New biofilm takes the composition of its neighbours.
(a) Share of species A (point-model species 1; B is species 2; blue: A, red: B) and (b) living volume fraction $\sum_i \bar{\phi}_i = \phi_A + \phi_B$, both shown where the biofilm $\phi_{AB} > 0.5$.
(c) Nutrient c, held at 1 in the bottom layer $y < -0.75$ mm and consumed by the living biomass. (d) Antibiotic a, held at 1 on the outer shell from $T^* = 1/3$ ($\alpha_0^* = 100$). One octant ($x, y, z > 0$) is removed to show the inside.
Constants: $\beta = 0.02$ mm^2/T^*, $r = 10$ mm/T^*, $K = 1$, $k_a = 10^{-1}/T^*$, $d = 10^0$ mm^2/T^* for c and a, $g\,l^2/d = 1$ for c and a, $\phi_{\max} = 10^{-1}$, $\alpha_0^* = 100$ from $T^* = 1/3$.
Sources: β, r, k_a, d, g from Klempt et al. 2024, Table 2, converted to the 2 mm cube; α_0^* and the onset from Klempt et al. 2026, case 4. $T^* = 1$ is the duration of the point model's data (Fritsch et al. 2025).

Figure 2: Separate start, $T^* = 1$: (a) $\phi_1/(\phi_1 + \phi_2)$, (b) $\sum_i \bar{\phi}_i$, (c) c, (d) a. Python, 64^3.

- A (point-model species 1) in the seed, B (species 2) in four blocks at the z edges; nutrient from the bottom, antibiotic from all outer faces from $T^* = 1/3$ ($\alpha_0^* = 100$)
- Newly reached cells take the composition of the biofilm they grow from; the antibiotic kills B and, where the nutrient is low, A

Mixed example with antibiotic



Coupled growth model: Klempt et al. 2024, Eqs. 34–36, front term on ($v = 10$ mm/T^*), with the point model of Klempt et al. 2026 (Fritsch et al. 2025, Case I, true parameters) at every point; 2 mm cube, 64^3 finite volumes, TVD front term.
Mixed start: species A and B in equal amounts together in the centre seed. New biofilm takes the composition of its neighbours.
(a) Share of species A (point-model species 1; B is species 2; blue: A, red: B) and (b) living volume fraction $\Sigma_i \bar{\phi}_i$, $\bar{\phi}_i = \phi_i \omega_i$, both shown where the biofilm $\bar{\phi}_\Sigma > 0.5$.
(c) Nutrient c, held at 1 in the bottom layer $y < -0.75$ mm and consumed by the living biomass. (d) Antibiotic a, held at 1 on the outer shell from $T^* = 1/3$ ($a_0^* = 100$). One octant ($x, y, z > 0$) is removed to show the inside.
Constants: $\beta = 0.02$ mm^2/T^*, $r = 10$ mm/T^*, $K = 1$, $k_a = 10^{-3}/T^*$, $d = 10^3$ mm^2/T^* for c and a, $g\,l^2/d = 1$ for c and a, $\phi_{\min} = 10^{-3}$, $\alpha_0^* = 100$ from $T^* = 1/3$.
Sources: β, r, k_a, d, g from Klempt et al. 2024, Table 2, converted to the 2 mm cube; α_0^* and the onset from Klempt et al. 2026, case 4. $T^* = 1$ is the duration of the point model's data (Fritsch et al. 2025).

Figure 3: Mixed start, $T^* = 1$: (a) $\phi_1/(\phi_1 + \phi_2)$, (b) $\sum_i \bar{\phi}_i$, (c) c, (d) a. Python, 64^3.

- A and B start together in the seed with equal amounts; same nutrient and antibiotic as before
- The antibiotic acts where $c^* = c_0^* c$ falls below $\alpha^* b_i$: the nutrient field decides where it kills

Five examples on the finest grid

Table 1: Seed means at $T^* = 1$ (Python, 128^3, TVD front term).

Example	$\sum_i \bar{\phi}_i$	$\phi_1/(\phi_1+\phi_2)$	c	$\alpha - 1$	von Mises [Pa]
Separate	0.965	0.978	0.539	9.32×10^{-4}	6.18×10^{-4}
Separate, antibiotic	0.147	0.951	0.910	9.33×10^{-4}	6.05×10^{-4}
Mixed, equal amounts	0.960	0.848	0.578	9.07×10^{-4}	4.87×10^{-4}
Mixed, equal amounts, antibiotic	0.038	0.613	0.970	9.05×10^{-4}	4.26×10^{-4}
Mixed (0.25, 0.30), antibiotic	0.037	0.511	0.971	9.04×10^{-4}	4.29×10^{-4}

Constants: Klempt et al. (2024), Table 2, converted; Fritsch et al. (2025), case I; $\alpha_0^* = 100$ from $T^* = 1/3$ (Klempt et al. 2026, case 4).

- The antibiotic lowers the living fraction of the seed from 0.965 to 0.147 (separate) and from 0.960 to 0.038 (mixed)
- $\alpha - 1$ hardly changes between examples (Eq. 36 is unchanged): the composition reaches the stress through c and the front term

The front term is an advection

$$\mathrm{Pe} = \frac{r\,h}{2\beta} \approx 31$$

on 16^3 ($r = 10\,\mathrm{mm}/T^*$, $\beta = 0.02\,\mathrm{mm}^2/T^*$). With a central gradient ϕ_{3D} oscillates and the element stops after two increments; a smaller time step does not help.

Constant added diffusion $rh/4$

First order. Stable in the element, but the stress on 16^3 is about a quarter of the converged value.

Lax–Wendroff term with an extremum sensor

Used in the element: second order away from extrema, ϕ_{3D} stays in $[0, 1]$; on 16^3 it matches Python with the same form.

TVD upwinding (MUSCL, minmod)

Used in Python for the main results; converged from 64^3 on.

Growth converges for every scheme, stress only for second-order ones

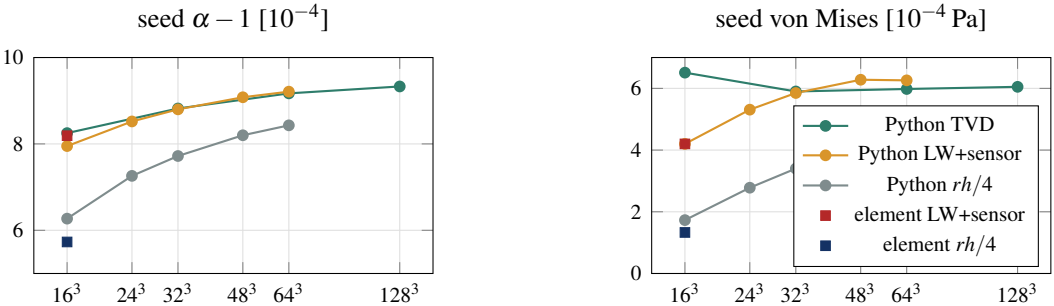


Figure 4: Seed means of the separate example with antibiotic at $T^* = 1$ against the grid ($h = 2\,\mathrm{mm}/n$). Element: your element with the agreed coupling (ANSYS), 16^3 so far.

What the comparison found in the element and in Python

Surface points of the element

ϕ at the surface points was an inverse-distance mean that also took inner points; at a wall with $\phi = 1$ it came out low and ϕ leaked through the wall. Taking the nearest Gauss point reduced the leak to a quarter.

Neighbour selection

The search keeps the first 30 points in element order; on 16^3 and 24^3 the stencils became asymmetric inside the domain. Selecting by distance makes them symmetric to 10^{-15}.

Column sums of the NEM Laplacian

The rows sum to zero, the columns do not, so the operator does not conserve ϕ with equal weights. A conservative form removes the rest of the leak.

Python wall gradient

My Python version took a one-sided gradient in the wall cells for the front term; corrected to the zero-flux mirror. After both corrections the codes agree on 8^3 within 1 to 5 %.

Growth around rigid bars (Klempt 2024, Sect. 4.3)

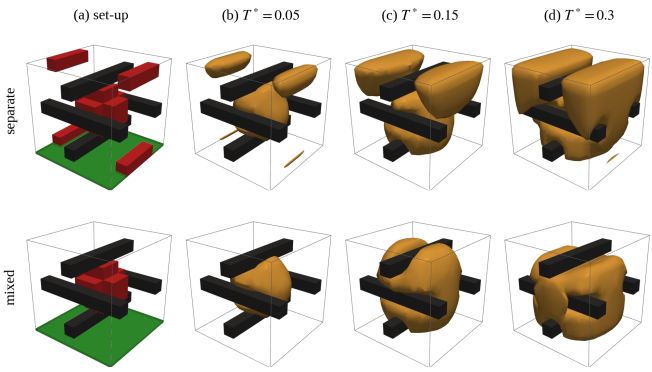


Figure 5: $\phi_{3D} \geq 0.5$ around four rigid bars, separate (top) and mixed (bottom) start; set-up after Klempt et al. (2024), Fig. 9.

- Four bars after Fig. 9 of the paper (positions read off the sketch), no biofilm or nutrient inside, frictionless contact
- The biofilm grows around the bars and closes behind them, as in Fig. 10
- The bars restrain the swelling: seed stress about ten times higher than without bars

The maze of Fig. 11 with the coupling

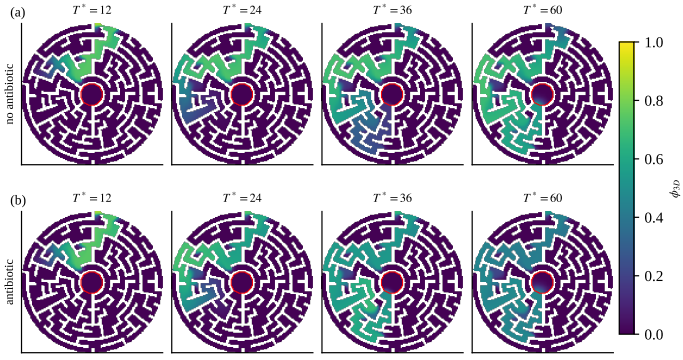


Figure 6: ϕ_{3D} in the maze of Klempt et al. (2024), Fig. 11 (560 µm across, 2D, $h = 4$ µm, Table 6 constants), without (a) and with (b) antibiotic held at the entrance from $T^* = 20$. Nutrient held in the centre (red), colony at the entrance (top).

The biofilm follows the corridors, also into dead ends, and comes within about 40 µm of the source by $T^* = 60$; with the antibiotic it stays about 130 µm away.

Questions and next steps

Questions for you

1. How was the front term stabilised in the runs of Klempt 2024? The cell Péclet number is about 25 on the paper's mesh.
2. Can I keep the three element fixes (nearest Gauss point at the surface, conservative Laplacian, neighbours by distance)?
3. Is a stabilised front term acceptable in the thesis, stated as a numerical treatment of Eq. 34?

Next steps

- Element runs on 24^3 with the Lax–Wendroff form for the convergence table
- Chapter 4 rewritten with these results by 13 November
- Thesis PDF by 14 December, colloquium on 16 December
- A joint paper on the coupling and its verification

References

F. Klempt, M. Soleimani, P. Wriggers, P. Junker. A Hamilton principle-based model for diffusion-driven biofilm growth. *Biomech. Model. Mechanobiol.* 23 (2024) 2091–2113.

F. Klempt, H. Geisler, M. Soleimani, P. Junker. A continuum multi-species bacterial growth model with a novel interaction scheme. *Arch. Appl. Mech.* 96 (2026) 164.

L. Fritsch, H. Geisler, J. Grashorn, F. Klempt, M. Soleimani, M. Broggi, P. Junker, M. Beer. Bayesian updating of constitutive parameters under hybrid uncertainties with a novel surrogate model applied to biofilms. arXiv:2512.15145 (2025).

Appendix A: notation

$\phi_{3D},\ \alpha$	biofilm phase field and growth ($\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$), Klempt et al. 2024
$c,\ a$	nutrient and antibiotic fields (first and second nutrient field of the element)
$\beta,\ k_\alpha,\ r,\ k$	diffusion of ϕ_{3D}, growth rate, front speed, Monod constant (Eqs. 34, 36)
$d,\ g$	diffusion and uptake of c and a (Eq. 35)
$\phi_i,\ \psi_i,\ \phi_0$	volume fraction and living fraction of species i, void (Klempt et al. 2026)
$\bar{\phi}_i = \phi_i\psi_i$	living volume fraction of species i; $\sum_i \bar{\phi}_i$ enters the consumption
$c^* = c_0^* c,\ \alpha^* = \alpha_0^* a$	nutrient and antibiotic seen by the point model (not the growth α)
$T^*,\ s$	normalised time; point-model time per unit T^* ($s = 0.1$)

Appendix B: equations and constants

$$\dot{\phi}_{3D} = \beta \nabla^2 \phi_{3D} + k_\alpha \alpha - r \tfrac{c}{k+c} \mathbf{n}_c \cdot \nabla \phi_{3D} \qquad \text{(Eq. 34 with the front term)}$$

$$\dot{c} = d \nabla^2 c - g \, \phi_{3D} \textstyle\sum_i \bar{\phi}_i \, c, \qquad \dot{a} = d \nabla^2 a - g \, \phi_{3D} \textstyle\sum_i \bar{\phi}_i \, a \qquad \text{(Eq. 35; } g\phi \text{ printed is a misprint)}$$

$$\dot{\alpha} = k_\alpha \, \phi_{3D} \qquad \text{(Eq. 36, unchanged)}$$

Point model of Klempt et al. 2026 at every Gauss point as published, with $\sum_i \phi_i + \phi_0 = 1$.

β	0.02 mm^2/T^*	Klempt 2024 Table 2, converted to the 2 mm cube (assumption)
d, g	10^8 mm^2/T^*, $g\ell^2/d = 1$	Table 2 (fields quasi-steady within $4 \times 10^{-8}\, T^*$)
r, k, k_α	10 mm/T^*, 1, 10^{-3} 1/T^*	Table 2
E, ν	10 Pa, 0.49	Table 2; void floor $f = 10^{-3}$ (converged)
point model, c_0^*, s	case I, 100, 0.1	Fritsch et al. 2025
α_0^*, onset	100 from $T^* = 1/3$	Klempt et al. 2026 case 4
$\phi_{\min}$	0.001	not in a paper; converged (assumption)